

JKBOSE 2026

CLASS 11TH



Cell Cycle And Cell Division

LECTURE-2

ZOOLOGY

DAVOOD SIR (CEO of NG)

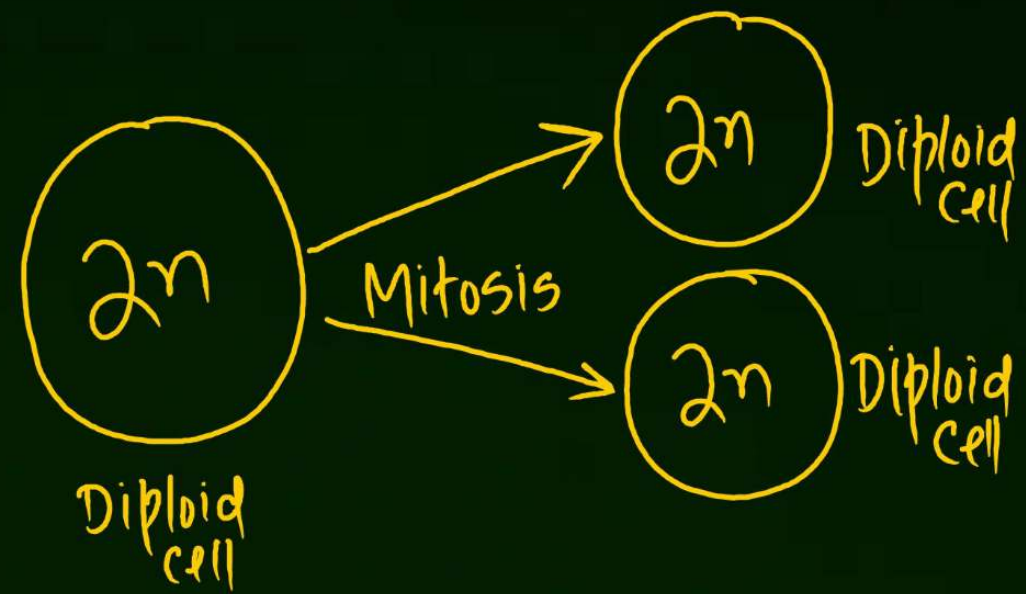
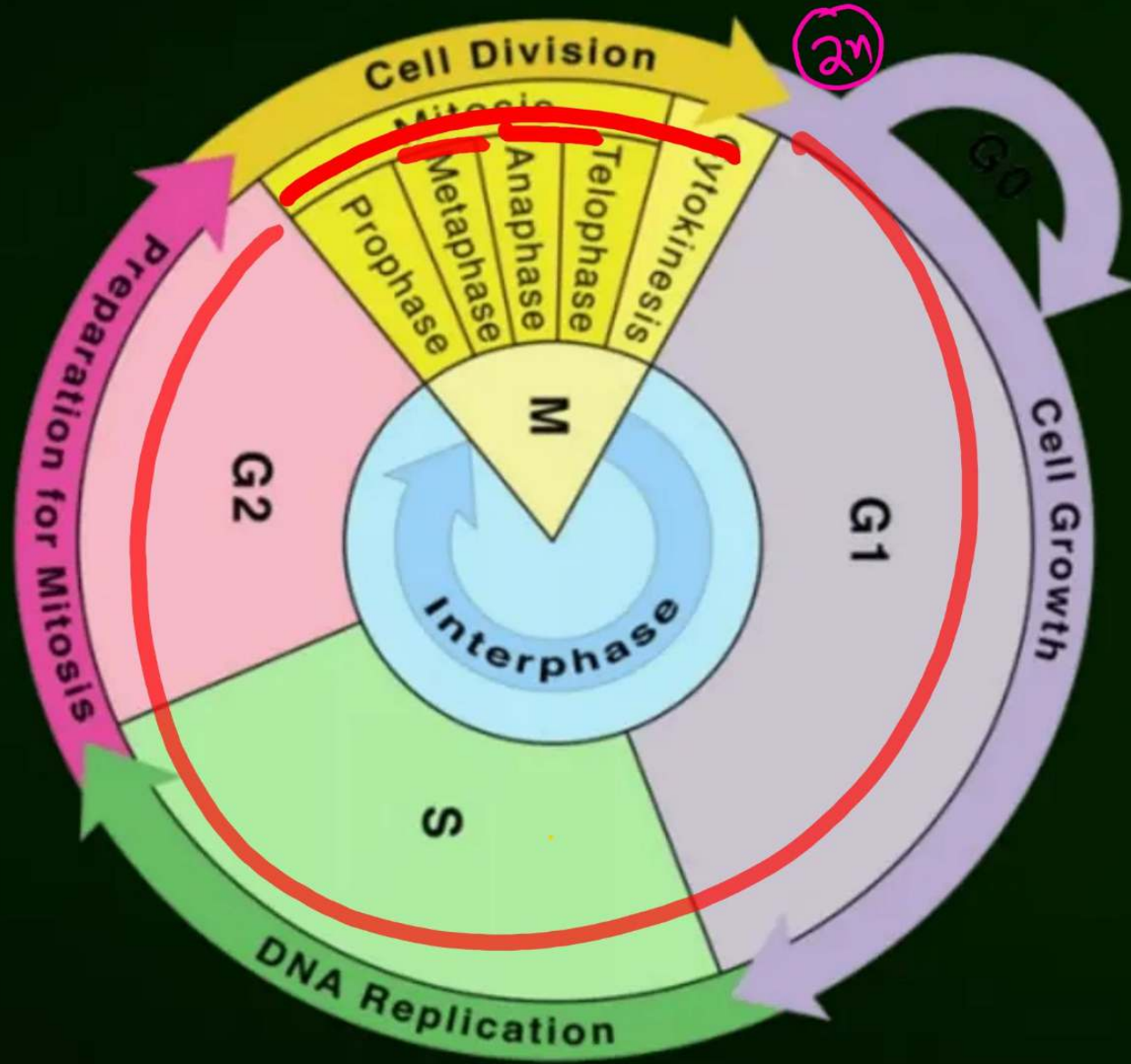


Physics Wallah

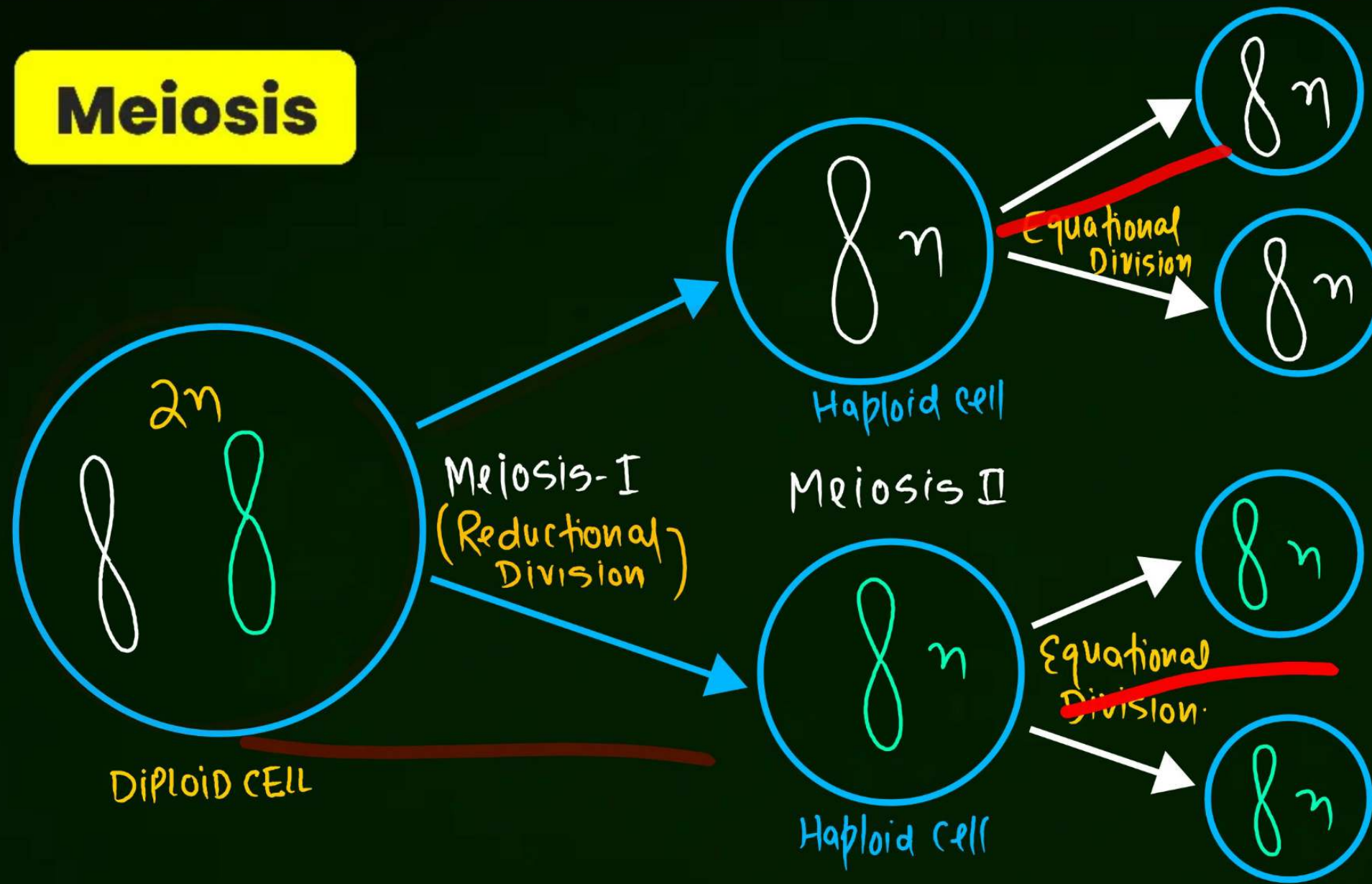
Topics To Be Covered

- 1 Meiosis (meiosis I)
- 2 Previous Year NEET MCQ'S

Previous Lecture



Meiosis

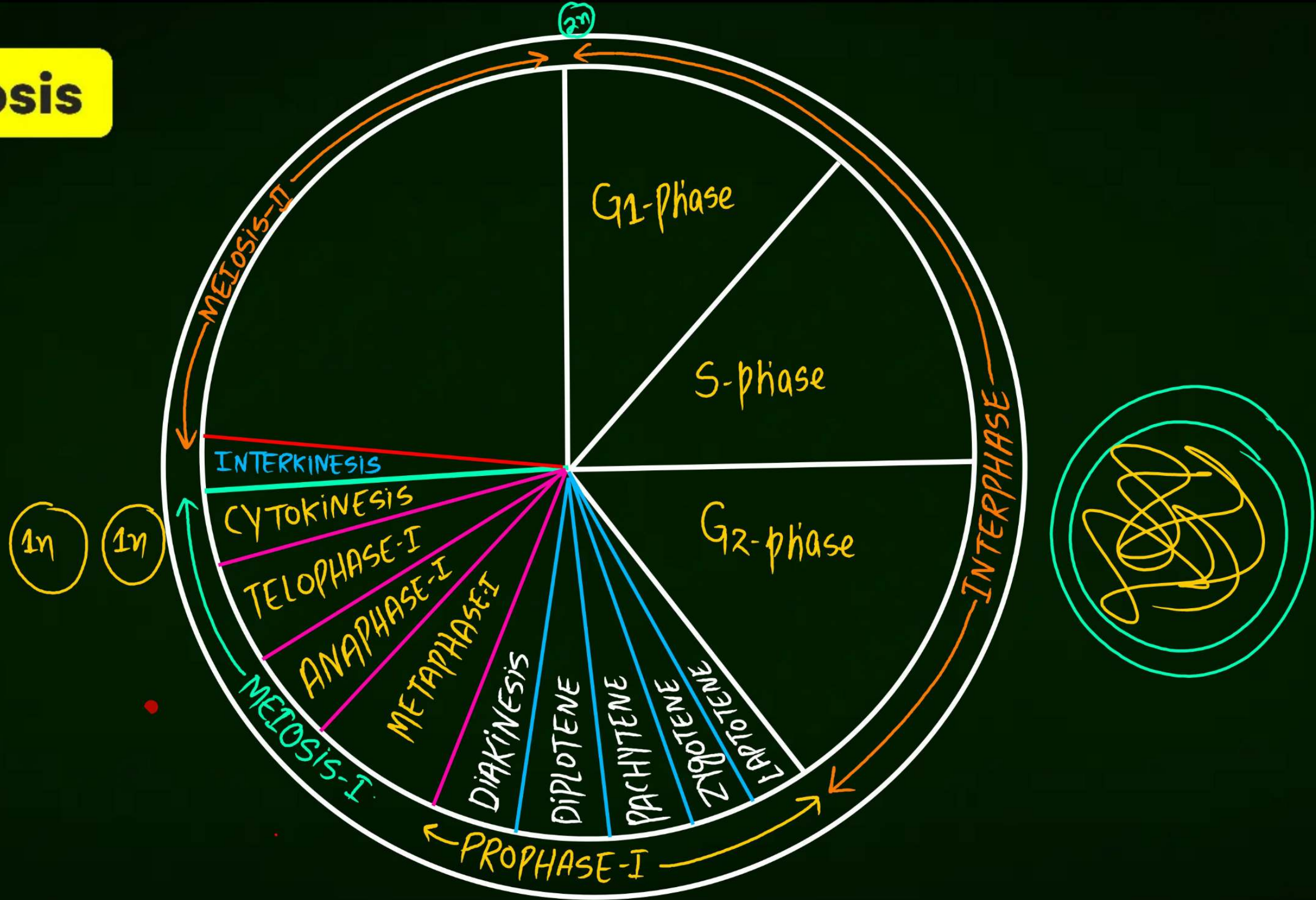


Meiosis

Meiosis is a specialized cell division that halves the chromosome number, producing haploid daughter cells. It occurs during gametogenesis in plants and animals, forming haploid gametes (sex cells).

- It involves two sequential divisions (meiosis I and II) but only one DNA replication.
- It begins after DNA replication in S phase, producing identical sister chromatids.
- It includes homologous chromosome pairing and recombination between non-sister chromatids.
- It ends with the formation of four haploid cells.

Meiosis

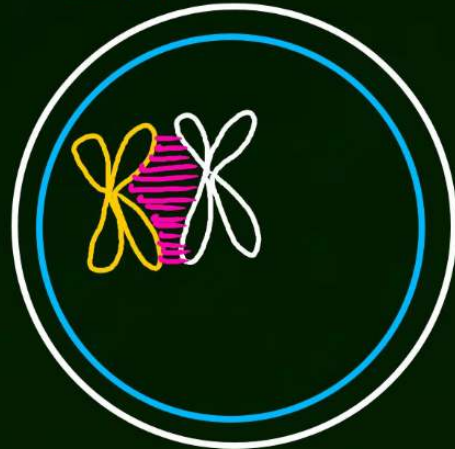


Prophase-I

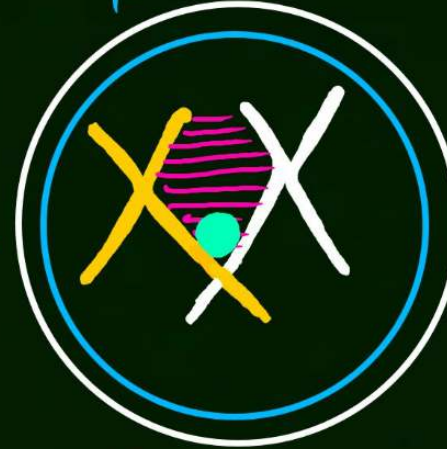
LAPTOTENE



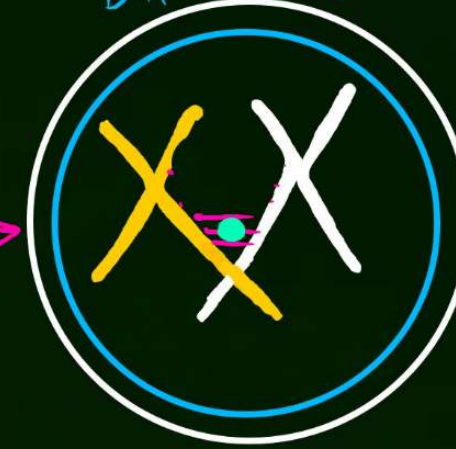
ZYGOTENE



PACHYTENE



DIPLLOTENE



DIKINESIS



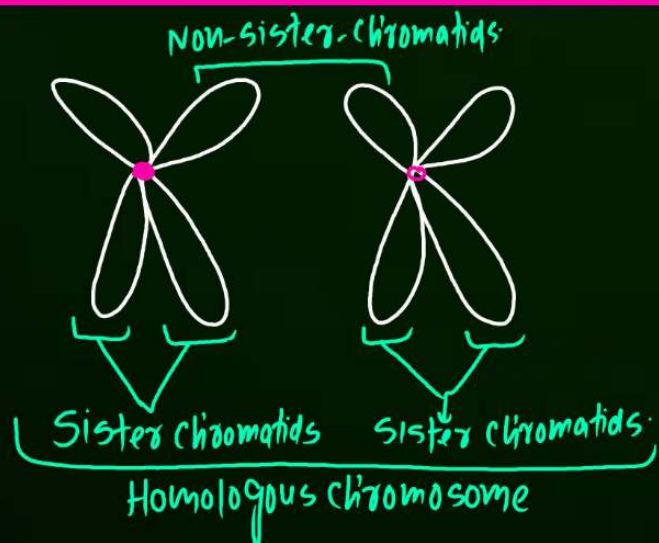
Compaction of Chromatin into chromosomes which are visible

- Homologous chromosomes start pairing due to Synaptonemal complex (protein) and it is called Synapsis and results in the formation of Bivalent or tetrad.

- Tetrad is clearly visible
- Formation of Recombinant nodule - this nodule contains an enzyme called Recombinase
- this helps in crossing over of non-sister chromatids of homologous chromosome.

- Dissolution of Synaptonemal complex.
- formation of chiasmata

- Terminalization of chiasmata
- Arrangement for spindle fibres.
- Nucleolus & Nuclear membrane disappears.



Question [NEET-2024]

Kaun Banayga Nabkaar



Match List I with List II.

List I

(Sub phases of prophase I)

A. Diakinesis

B. Pachytene

C. Zygotene

D. Leptotene

List II

(Specific characters)

I. Synaptonemal complex formation

II. Completion of terminalisation of chiasmata

III. Chromosome look like thin threads

IV. Appearance of recombination nodules

Choose the correct answer from the options given below.

1 A-IV, B-II, C-III, D-I



2 A-I, B-II, C-IV, D-III



3 A-II, B-IV, C-I, D-III



4 A-IV, B-III, C-II, D-I



Prophase I (meiosis I)

Prophase of the first meiotic division is typically longer and more complex when compared to prophase of mitosis. It has been further subdivided into the following five phases based on chromosomal behaviour, i.e.,

- a) Leptotene,
- b) Zygotene,
- c) Pachytene,
- d) Diplotene and
- e) Diakinesis.

Stages of Prophase I

- a) **Leptotene:-** The compaction of chromosomes continues throughout leptotene and becomes visible under light microscope
- b) **Zygotene:-** Chromosomes start pairing together and this process of association is called synapsis & occurs due to synaptonemal complex. Such paired chromosomes are called homologous chromosomes. The complex formed by a pair of synapsed homologous chromosomes is called a bivalent or a tetrad.
- c) **Pachytene:-** During this the four chromatids of each bivalent chromosome become clearly visible, forming tetrads. Recombination nodules appear as specific sites where crossing over occurs between non-sister chromatids of homologous chromosomes. Crossing over, mediated by the enzyme recombinase, involves the exchange of genetic material between homologous chromosomes. By the end of this stage, the chromosomes remain attached at the points where crossing over has taken place.

Stages of Prophase I

(d). Diplotene:- Diplotene begins with the dissolution of the synaptonemal complex. Recombined homologous chromosomes of bivalents start to separate, remaining attached only at crossing over sites, forming X-shaped structures called chiasmata. In oocytes of vertebrates, diplotene can last for months to years.

(e). Diakinesis:- This is marked by terminalisation of chiasmata. Chromosomes are fully condensed and the meiotic spindle is assembled to prepare the homologous chromosomes for separation. The nucleolus disappears and the nuclear envelope also breaks down.

Question [NEET-2023]

Kaun Banayga Nabkaar



The dissolution of synaptonemal complex occurs during

1 Diplotene. ✓

2 Diakinesis

3 Laptotene

4 Zygotene

Question [NEET-2024]

Kaun Banayga Nabkaar

Given below are two statements.

Statement I: Chromosomes become visible under light microscope during leptotene stage. *T*

Statement II: The beginning of diplotene stage is recognised by dissolution of synaptonemal complex. *T*

In the light of the above statements, Choose the correct answer from the options given below.

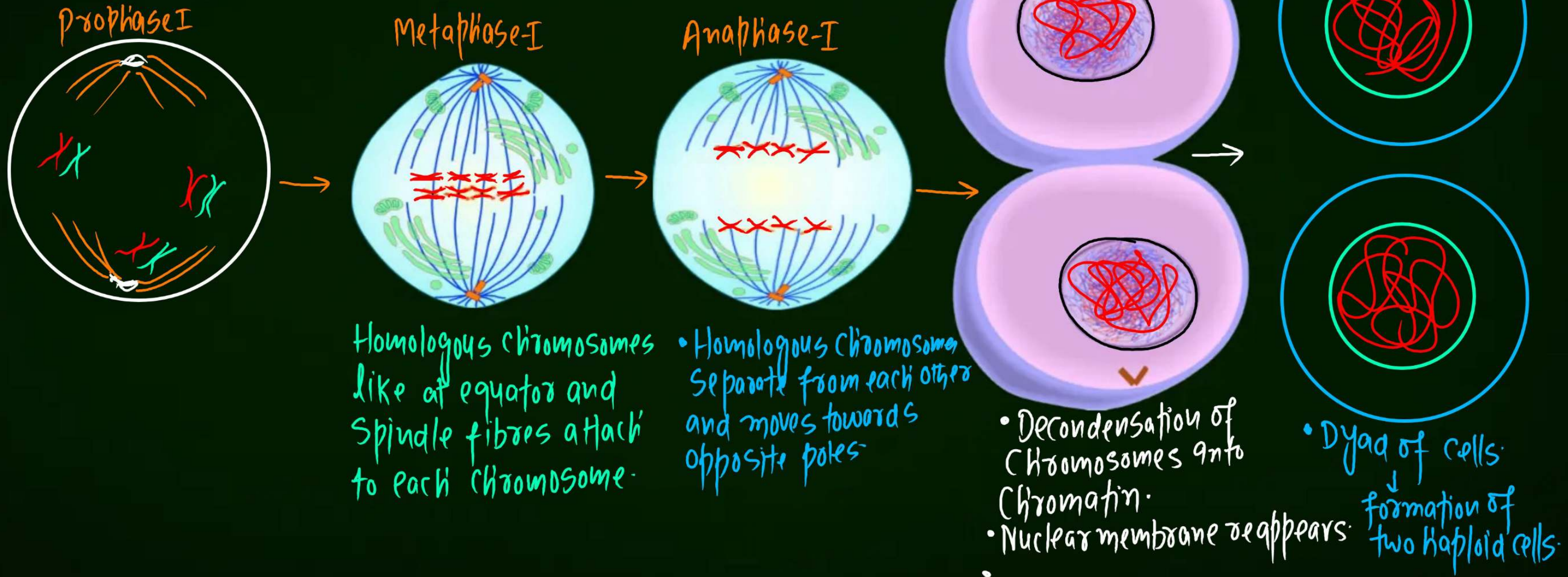
1 Both Statement I and Statement II are true. ✓

2 Both Statement I and Statement II are false

3 Statement I is true but Statement II is false

4 Statement I is false but Statement II is true

MEIOSIS-I



MEIOSIS-I

2. METAPHASE:- The bivalent chromosomes align on the equatorial plate. The microtubules from the opposite poles of the **spindle attach to the kinetochore** of homologous chromosomes.

3. ANAPHASE:- The homologous chromosomes separate, while sister chromatids remain associated at their centromeres.

4. TELOPHASE:- The nuclear membrane and nucleolus reappear, cytokinesis follows and this is called as dyad of cells.

Interkinesis

The stage between the two meiotic divisions is called **interkinesis** and is generally short lived. There is no replication of DNA during interkinesis. Interkinesis is followed by prophase II of meiosis II.

Regarding meiosis, which of the statement is incorrect?

- 1** There are two stages in meiosis, meiosis-I and II
- 2** DNA replication occurs in S phase of meiosis-II. ✓
- 3** Pairing of homologous chromosomes and recombination occurs in meiosis-I
- 4** Four haploid cell are formed at the end of meiosis-II .

Thank

You